Cryptography project

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Cryptography Tool

RAIL FENCE CIPHER

Rail fence (zigzag) depends on :

1. Depth (number of rows)

2. Read from left to right row by row

Plain text =Hello word , Depth = 3

h				o				d
	e		l		w		r	
		l				o		

h				o				d
	e		l		w		r	
		l				o		

The result : hodelwrlo

```python
def rail_fence_encryption(plaintext, depth):
    rail = ['' for i in range(depth)]
    # Suppose depth = 3
    #rail = [' ', ' ', ' ']

    direction_change = 1  # Default direction is dawn
    row = 0

    for char in plaintext:
        rail[row] += char
        if row == 0:
            direction_change = 1
        elif row == depth - 1:
            direction_change = -1
        row = row + direction_change
    # Join the rails to form the ciphertext
    encrypted_text = ''.join(rail)

    return encrypted_text

print (rail_fence_encryption("helloword" , 3 ))
```

Declaration of list contain strings the same number as the depth each element will be a row

Direction 1 mean dawn
Direction -1 mean up
Start from row index 0

In the first row direction = 1 to go dawn
When reach depth -1 (the last row) let the direction be -1 to go up

Join all the characters to make it as one string

1)Change number one : Perform the ceaser cipher at each rows

h				o				d
	e		l		w		r	
		l				o		

Assume key = 3
C=(h,3) = h+3 % 26
h=7
7+3 % 26 =10
10 = k
So h →k

P= hod c = (P,K) P+K % 26 = C → krg

P= elwr c = (P,K) P+K % 26 = C → hozu

P= lo c = (P,K) P+K % 26 = C → or

The code for ceaser cipher :

```python
def caesar_cipher_standard(text, shift):
    text = text.lower()  # Convert all characters to lowercase
    encrypted_text = ""
    for char in text:
        if char.isalpha():
            base = ord('a')
            shifted_char = chr(((ord(char) - base + shift) % 26) + base)
            encrypted_text += shifted_char
        else:
            print("error ! only alphabets ")
    return encrypted_text
```

Ord ( ) is a function that take a character and return its Ascii code
Ord(a)=97
Assume shift =3
If the character was b : we take the ord(b) = 98 then 98-97 = 1 + 3 = 4
4 which is e so char (4 + 97) = (101) = e

2)Change number two : apply Randomization to make it more complicated and harder to break

 # Make a random order of the rows

Assume the depth is 3 so the order of reading the rows is : 1 , 2 , 3
After apply the random method the order of reading The rows can be : 2 , 3 , 1 / 3 , 1 , 2 / 2 , 1 , 3 ……

Using the random. Shuffle function

The code for the random method :

```
def rail_fence_encrypt(plaintext, depth, ceaser_key, direction='left'):
    rail = ['' for _ in range(depth)]
    direction_change = 1  # Default direction is upward
    row = 0
    for char in plaintext:
        rail[row] += char
        if row == 0:
            direction_change = 1
        elif row == depth - 1:
            direction_change = -1
        row += direction_change
    # Shuffle the order of rails
    print (rail)
    random.shuffle(rail)
    print(rail)
```

Rail = [hod , elwr , lo] # the rows

random order of rows (change the row order)

Rail = [lo , hod , elwr] # new rows order

3)Change number three : make an option to read from left to right or from right to left

By adding one parameter that take only right or left :-
Left (default) : left to right
Right : from right to left

h				o				d
	e		l		w		r	
		l				o		

Right to Left

doh

rwle

ol

The code :

```
# Reverse the string if reading direction is specified as 'right'
if direction == 'right':
    rail[i] = rail[i][::-1]
```

If rail [i] =	h	e	l	l	o	w	o	r	d
	-9	-8	-7	-6	-5	-4	-3	-2	-1

Rail [i] [: : -1] # start from -1 and go backward

```python
import random
def caesar_cipher_standard(text, shift):
    text = text.lower()  # Convert all characters to lowercase
    encrypted_text = ""
    for char in text:
        if char.isalpha():
            base = ord('a')
            shifted_char = chr(((ord(char) - base + shift) % 26) + base)
            encrypted_text += shifted_char
        else:
            print("error ! only alphabets ")
    return encrypted_text

def rail_fence_encrypt(plaintext, depth , ceser_key ,  direction='left'):
    rail = [''  for i in range(depth)]
    # Suppose depth = 3
    #rail = [' ', ' ', ' ']

    direction_change = 1  # Default direction is upward
    row = 0

    for char in plaintext:
        rail[row] += char
        if row == 0:
            direction_change = 1
        elif row == depth - 1:
            direction_change = -1
        row = row + direction_change
```

```
        # Shuffle the order of rails
    print (rail)
    random.shuffle(rail)
    print(rail)
    # Apply Caesar cipher to each rail
    for i in range(depth):
        rail[i] = caesar_cipher_standard(rail[i], ceaser_key)

        if direction == 'right':
            rail[i] = rail[i][::-1]

    encrypted_text = ''.join(rail)
    return encrypted_text

plaintext = "helloword"
depth = 3
ceaser_key=1
encrypted_text = rail_fence_encrypt(plaintext, depth ,ceaser_key , "right")
print("Encrypted text:", encrypted_text)
```

The output :

```
order of rows : ['hod', 'elwr', 'lo']
order of rows sfter the random shuffle ['elwr', 'lo', 'hod']
Encrypted text : sxmfpmepi
```

The end

Thanks